

P1.4 Gravitational potential energy stores

Learning objectives

After this topic, you should know:

- what happens to the gravitational potential energy stores of an object when it moves up or down
- why an object moving up has an increase in its gravitational potential energy store
- why it is easier to lift an object on the Moon than on the Earth
- how to calculate the change in gravitational potential energy of an object when it moves up or down.

Worked example

A student of weight 300 N climbs on a platform that is 1.2 m higher than the floor. Calculate the increase in her gravitational potential energy store.

Solution

$$\text{Increase of } E_p = 300 \text{ N} \times 1.2 \text{ m} = 360 \text{ J}$$

Worked example

A 2.0 kg object is raised through a height of 0.4 m. Calculate the increase in the gravitational potential energy store of the object. The gravitational field strength of the Earth at its surface is 9.8 N/kg.

Solution

$$\begin{aligned} \text{Gain of } E_p &= \text{mass} \times \text{gravitational field strength} \times \text{height gain} \\ &= 2.0 \text{ kg} \times 9.8 \text{ N/kg} \times 0.4 \text{ m} \\ &= 7.8 \text{ J} \end{aligned}$$

Changes in gravitational potential energy stores

Every time you lift an object up, you do some work. Some of your muscles transfer energy from the chemical energy store in the muscle to the gravitational energy store of the object. In calculations we refer to the energy in this store as gravitational potential energy E_p .

The force you need to lift an object at constant velocity is equal and opposite to the gravitational force on the object. So the upward force you need to apply to it is equal to the object's weight. For example, you need a force of 80 N to lift a box of weight 80 N.

- When an object is moved upwards, the energy in its gravitational potential energy store increases. This increase is equal to the work done on it by the lifting force to overcome the gravitational force on the object.
- When an object moves down, the energy in its gravitational potential energy store decreases. This decrease is equal to the work done by the gravitational force acting on it.

The work done when an object moves up or down depends on:

- 1 how far it is moved vertically (its change of height)
- 2 its weight.

Using work done = force applied $\times$ distance moved in the direction of the force:

$$\begin{array}{l} \text{change in object's gravitational} \\ \text{potential energy store} \\ \text{(joules, J)} \end{array} = \begin{array}{l} \text{weight} \\ \text{(newtons, N)} \end{array} \times \begin{array}{l} \text{change of height} \\ \text{(metres, m)} \end{array}$$

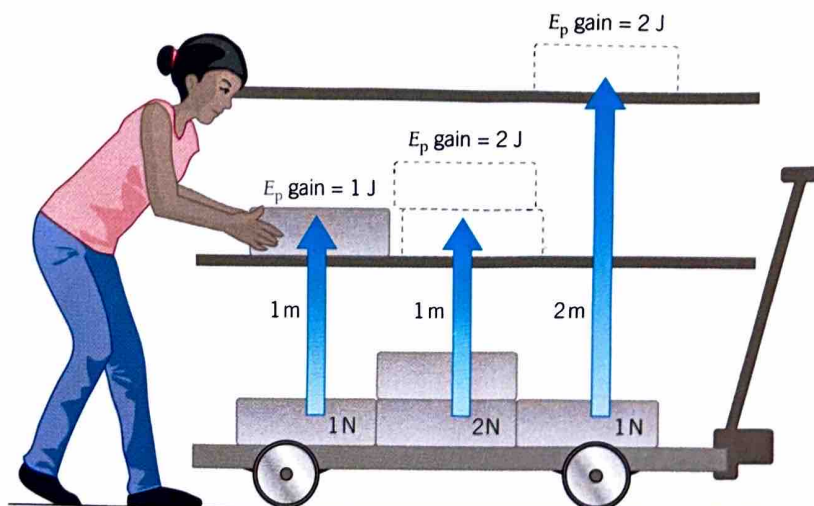


Figure 1 Using joules

Gravitational potential energy stores and mass

Astronauts on the Moon can lift objects much more easily than they can on the Earth. This is because the gravitational field strength on the Moon's surface is only about a sixth of the gravitational field strength on the Earth's surface.

You have previously learnt that the weight of an object in newtons is equal to its mass $\times$ the gravitational field strength. So, when an object is lifted or lowered, the change to its gravitational potential energy store is equal to its weight $\times$ its change of height. Therefore:

$$\begin{array}{l} \text{change of} \\ \text{gravitational} \\ \text{potential} \\ \text{energy store, } \Delta E_p \\ \text{(joules, J)} \end{array} = \begin{array}{l} \text{mass, } m \\ \text{(kilograms, kg)} \end{array} \times \begin{array}{l} \text{gravitational} \\ \text{field strength, } g \\ \text{(newtons per} \\ \text{kilogram, N/kg)} \end{array} \times \begin{array}{l} \text{change of} \\ \text{height, } \Delta h \\ \text{(metres, m)} \end{array}$$

- 1 a Describe the changes to the energy stores of a ball when it falls and rebounds without regaining its initial height. [4 marks]
- b When a ball of weight 1.4 N is dropped from rest from a height of 2.5 m above a flat surface, it rebounds to a height of 1.7 m above the surface.
 - i Calculate the total energy lost from the ball's energy stores when it reaches this maximum rebound height. [2 marks]
 - ii Name two causes of the energy transfer. [2 marks]
- 2 A student of weight 450 N steps on a box of height 0.2 m.
 - a Calculate the increase in her gravitational potential energy store. [1 mark]
 - b Calculate the work done by the student if she steps on and off the box 50 times. [1 mark]
- 3 a A weightlifter raises a steel bar of mass 25 kg through a height of 1.8 m. Calculate the change to the gravitational potential energy store of the bar. The gravitational field strength at the surface of the Earth is 9.8 N/kg. [2 marks]
- b The weightlifter then lowers the bar by 0.3 m and drops it so it falls to the ground. Assume that air resistance is unimportant. Calculate the change to its gravitational potential energy store in this fall. [2 marks]
- 4 You use energy when you hold an object stationary in your outstretched hand. Suggest what happens to the energy that must be supplied to your muscles to keep them contracted.  [3 marks]

Study tip

Watch out for objects going up a slope. To calculate the increase in their gravitational potential energy stores, you need the vertical height gained, *not* the distance along the slope.

Study tip

In physics, the greek letter delta (Δ) is used to represent the phrase 'change in'. For example, Δh can be used in place of 'change in height'.

Stepping up

Measure your mass in kilograms using floor scales.

Step on and off a sturdy box or low platform.

Measure the height of the box.

Use the equation $\Delta E_p = m g \Delta h$, where $g = 9.8 \text{ N/kg}$, to calculate the energy transferred to your gravitational potential energy store when you stepped on the box.

Safety: Make sure the box is secure and that you feel comfortable to do this.



Key points

- The gravitational potential energy store of an object increases when it moves up and decreases when it moves down.
- The gravitational potential energy store of an object increases when it is lifted up because work is done on it to overcome the gravitational force.
- The gravitational field strength at the surface of the Moon is less than on the Earth.
- The change in the gravitational potential energy store of an object is $\Delta E_p = m g \Delta h$